

Mater — End-to-End System Outline

Detection + Dispatch/Claim Layer · Sep 24, 2026 · Companion to the Vehicle Detection v1 PRD

0. One-line scope

Camera detects a violation on a property -> system builds a defensible evidence record -> confirmed job routes to the one towing company contracted to that property -> a driver at that company claims and resolves it. One product, from first frame to closed job.

I. Core entities (data model, end to end)

Existing (from original PRD / mater-demo repo)

- **orgs, org_members** — tenancy, RLS by org
- **regions** — a property (has GeoJSON polygon, time limits, auto-ticketing flag)
- **cameras** — one per physical camera, config (lines, spaces, masks, thresholds)
- **spaces** — a monitored parking space (polygon, label, reserved flag)
- **scan_events** — the crossing/detection timeline (photos, timestamps), extended with direction, camera_id, track_id
- **space_events** — occupancy state changes (occupied/vacated)
- **presences** — one open row per vehicle currently on the lot (plate, entry event, exit event, timestamps)

New (this session's scope)

- **towing_companies** — a licensed tow operator entity
- **property_company_contracts** — the property <-> company authorization (one active company per property, or an ordered fallback list); this is the legal fact that gates everything downstream
- **drivers / trucks** — belong to exactly one towing_company; a driver may be assigned to one or several properties under that company's contracts
- **violation_rules** — per-property config: permit list and/or time-limit and/or restricted-zone, plus the evidentiary buffer duration for that property
- **violations** — one row per candidate violation: rule matched, first_detected_at, evidence pointers, state (pending -> dismissed | confirmed)
- **jobs** — one row per confirmed violation once it's dispatchable: state (open -> claimed -> resolved:towed | resolved:lost), company_id, claimed_by, timestamps

II. Full pipeline, one violation start to finish

- 1 **Ingest** — camera (RTSP / file / YouTube-resolved-via-yt-dlp) streams frames at configured fps (default 5); configured mask regions (e.g. timestamp overlay) applied before detection; stream drops trigger auto-reconnect with backoff.
- 2 **Detect** — YOLO (Ultralytics, pretrained COCO) finds car/truck/bus/motorcycle boxes; detections below confidence threshold (default 0.4) dropped.

- 3 **Track** — ByteTrack assigns a stable track_id to each vehicle for as long as it's in view.
- 4 **Entry event** — track crosses a configured counting line inward -> entry scan_event fires -> opens a presence row for that vehicle.
- 5 **Exit event** (parallel, ends the story later) — track crosses the line outward -> exit scan_event fires -> closes the matching open presence (matched by plate within the region; unmatched exits still logged, not dropped). Crossings jitter-debounced (ignore back-and-forth within 2 seconds).
- 6 **Occupancy tracking** (parallel path, for space-based rules) — vehicle box overlaps a space polygon above threshold (default 40%) and holds for 30 seconds -> space marked occupied, linked to the track/plate; reverse for vacated. This 30s hold exists purely to suppress noise from passing cars — separate concept from the evidentiary buffer in step 8.
- 7 **Plate read** — plate detection + OCR (fast-alpr) runs on the vehicle crop every frame the track is visible; the single highest-confidence read (threshold default 0.8) is kept and attached to the track/presence. No qualifying read -> event proceeds with no plate.
- 8 **Rule evaluation** — for every tracked vehicle, check that property's violation_rules: permit-list rule (plate not on the authorized list for this space/lot); time-limit rule (dwell time exceeds the configured limit); restricted-zone rule (vehicle detected in a disallowed area). A match opens a violations row: first_detected_at, rule type, property_id, camera_id, track_id, plate, evidence pointer. State = pending.
- 9 **Evidentiary buffer** — the violation holds in pending for a per-property configurable window (default ~10 min). Through that window, the system keeps capturing periodic timestamped snapshots — not just start/end — building a continuous, dispute-proof record that the vehicle was there, unmoved, out of compliance, the whole time.
- 10 **Resolution check (during buffer)** — if the condition clears before the buffer expires (vehicle exits, permit appears, time limit resets), the violation is marked dismissed. No job is created. Evidence is retained but archived, not surfaced.
- 11 **Confirmation** — buffer expires with the condition still unresolved -> violation state flips to confirmed, confirmed_at recorded. This happens automatically; no human reviews or approves it (default assumption validated for Providence-style jurisdictions — kept as a per-property toggle for stricter jurisdictions/contracts).
- 12 **Job creation** — system looks up property_company_contracts for this property, resolves the authorized company (or first in the fallback order), and creates a jobs row in state open, carrying the full evidence bundle from the violation.
- 13 **Queue visibility** — every driver belonging to that company, for every property they're assigned to, sees this job pooled alongside every other open job across all their assigned properties — no system-side matching or ranking, just a shared list.
- 14 **Driver review** — driver sees plate, evidence snapshots, property/location, violation type, and confirmation time in the queue item.
- 15 **Claim** — driver taps "Accept." This is a single atomic write: first successful claim wins, job moves to claimed with claimed_by and claimed_at set, and it disappears (or grays out) on every other driver's screen in real time.
- 16 **Drive to location** — no in-app routing/ETA logic for MVP; the driver just goes. No urgency SLA — the evidentiary buffer already filtered out short-dwell vehicles, so there's no rider-cancellation-style time pressure.
- 17 **Arrival outcome** — driver marks one of two terminal states: *Towed* (vehicle confirmed on hook/removed — job -> resolved:towed, presence closes with exit-reason towed, evidence bundle finalized and locked); *Lost* (vehicle already gone or condition resolved — job -> resolved:lost, presence closes with exit-reason left-before-tow, no penalty or reliability scoring, just logged).
- 18 **Dashboard reflects reality** — "On the lot now" list drops the vehicle either way; entry/exit log gets a finalized entry with the complete evidence timeline (first detection through resolution); space map updates if a monitored space was involved.

- 19 **Retention** — full record (rule matched, every snapshot, buffer duration, confirmation timestamp, claiming driver, outcome) kept for a per-property configurable retention period (default 30 days), exportable as the compliance/dispute-defense artifact.

III. Roles

Role	Sees / does
Property manager	Configures violation_rules and buffer duration for their property; views dashboard, on-the-lot-now, space map, camera health, full audit/export trail
Towing company admin	Manages driver/truck roster; sees company-wide open/claimed/resolved job stats across all properties they're contracted to
Driver	Sees pooled claim queue across their assigned properties; claims and resolves jobs; no visibility into properties they're not assigned to
System (detector + rule engine + buffer scheduler)	No human account; runs the pipeline above autonomously

IV. Legal/compliance facts baked into the architecture (not bolt-ons)

- A job can only ever be visible to the one company under contract for that property (property_company_contracts gates the queue at the data layer, not the UI).
- Rules are per-property, not global — because permits, time limits, and contract terms differ property to property.
- Auto-confirmation is a per-property toggle, defaulting to on (validated for Providence) but switchable to a human-approval step for stricter jurisdictions or nervous property managers.
- The evidentiary buffer's duration is per-property configurable, not hardcoded — different contracts may specify different grace periods.
- Retention period is per-property configurable, defaulting to 30 days, per the original PRD's privacy section.

V. Explicit non-goals (MVP)

- No billing, payroll, invoicing, or storage-lien processing — customers keep existing back-office software (ParkingSnap/Omadi/Autura-class tools) for that.
- No cross-company marketplace or bidding — fleet-to-property is a fixed contractual fact, never a live match.
- No driver reliability scoring or no-show penalties — “lost” is a terminal state, not a strike.
- No last-look re-verification before arrival — no urgency problem to solve for.
- No native mobile app requirement — a responsive web view suffices for both driver queue and dashboard.
- No gating the MVP launch on hitting the PRD's 85% plate-accuracy target — plate reading matters for permit-list matching and evidence quality, but it's a tuning target, not a go/no-go.
- No custom model training — pretrained YOLO/ByteTrack/fast-alpr only.

VI. Build order

- 1 Detector core (video input -> YOLO + ByteTrack -> line-crossing + occupancy zones) — validate against the Brighton livestream and recorded test clips.
- 2 Rule engine + evidentiary buffer state machine (pending -> dismissed | confirmed) — validate the confirm/dismiss logic in isolation before any dispatch code exists.
- 3 Job + claim queue (open -> claimed -> resolved), atomic claim mechanic, with one fake company/driver for testing.
- 4 Property manager dashboard (on-the-lot-now, entry/exit log with evidence, space map, job/audit export).
- 5 One real pilot: one property, one real contracted tow company, real camera(s), live for several weeks — measure violations surviving the buffer, time-to-claim, time-to-resolution, and whether the evidence held up in any real dispute.

Companion document to "Mater — PRD: Vehicle Detection v1 (Entry/Exit + Occupancy)." This outline extends that PRD's scope to include the dispatch/claim layer, which the original PRD listed as a later, separate PRD.